

$$\boxed{1} \boxed{0} . \boxed{1}$$

Contrast how an optical microscope and a transmission electron microscope work **and** contrast the limitations of their use when studying cells.

[6 marks]

[illegible]

Figure 6



[4 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

An environmental scientist investigated a possible relationship between air pollution and the size of seeds produced by one species of tree.

He was provided with a very large number of seeds collected from a population of trees in the centre of a city and also a very large number of seeds collected from a population of trees in the countryside.

Describe how he should collect and process data from these seeds to investigate whether there is a difference in seed size between these two populations of trees.

[5 marks]

[illegible]

END OF QUESTIONS